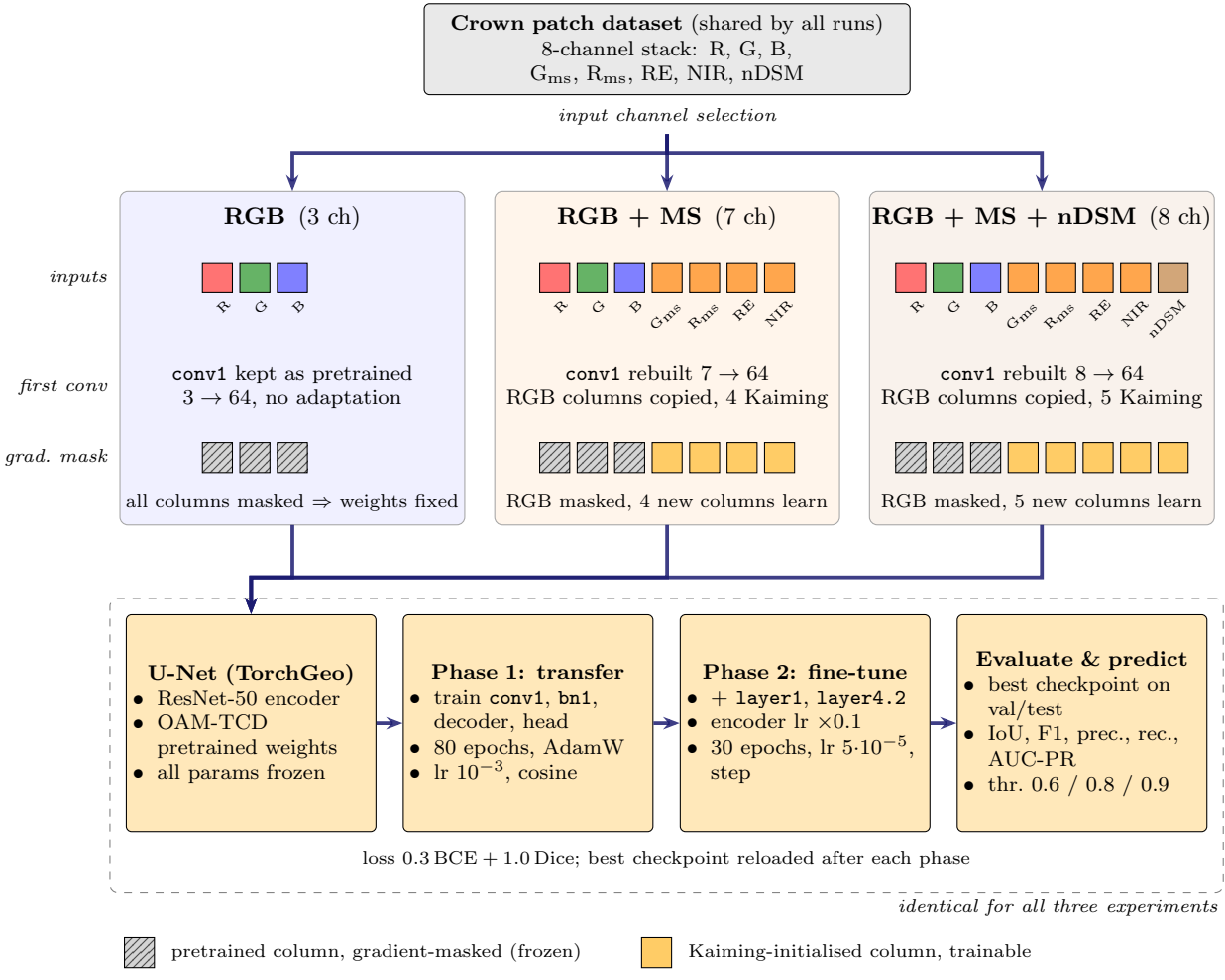

Crown patch dataset (shared by all runs)
8-channel stack: R, G, B,
Gms, Rms, RE, NIR, nDSM
input channel selection
RGB (3 ch)
RGB + MS (7 ch)
RGB + MS + nDSM (8 ch)
inputs
first conv
grad. mask
R G B
R G B Gms Rms RE NIR
R G B Gms Rms RE NIR nDSM
conv1 kept as pretrained
3 → 64, no adaptation
conv1 rebuilt 7 → 64
RGB columns copied, 4 Kaiming
conv1 rebuilt 8 → 64
RGB columns copied, 5 Kaiming
all columns masked ⇒ weights fixed
RGB masked, 4 new columns learn
RGB masked, 5 new columns learn
U-Net (TorchGeo)
ResNet-50 encoder
OAM-TCD pretrained weights
all params frozen
Phase 1: transfer
train conv1, bn1, decoder, head
80 epochs, AdamW
lr 10−3, cosine
Phase 2: fine-tune
+ layer1, layer4.2
encoder lr ×0.1
30 epochs, lr 5·10−5, step
Evaluate & predict
best checkpoint on val/test
IoU, F1, prec., rec., AUC-PR
thr. 0.6 / 0.8 / 0.9
loss 0.3 BCE + 1.0 Dice; best checkpoint reloaded after each phase
identical for all three experiments
pretrained column, gradient-masked (frozen)
Kaiming-initialised column, trainable